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Class: CS4323.Q12.CTTT

OS
LAB 2

CHECKLIST

1. Homework

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Your code	☒	☒	☒	☒
Results and discussions	☒	☒	☒	☒

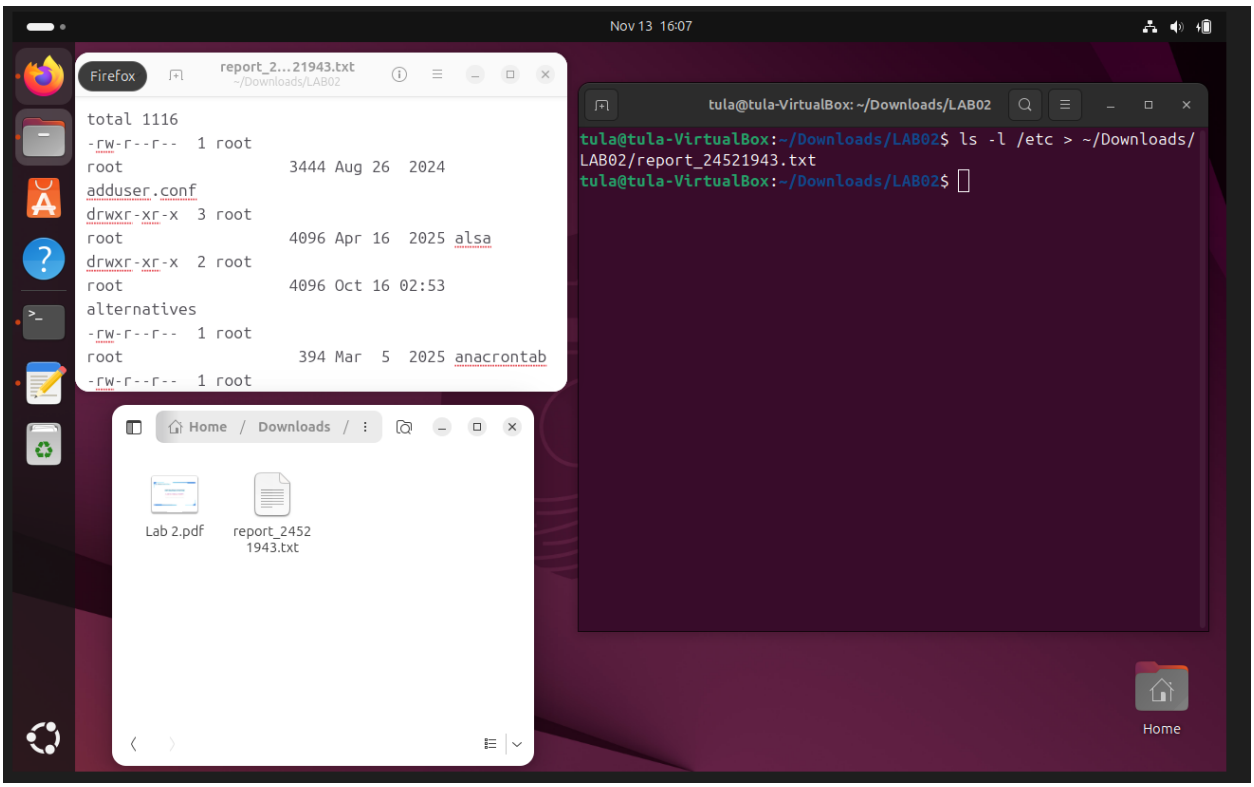
<u>Mark:</u>

2. Lab 2

1. List the contents of your /etc directory, including the ownership and permissions, and redirect the output to a file called report_<studentID>.txt within your home directory.

Answer:

Code: ls -l /etc > ~/Downloads/LAB02/report_24521943.txt



Explanation:

`ls -l /etc`

Lists all files in `/etc` with **permissions**, **owners**, **group**, **size**, and **timestamps**.

`>`

Redirects the output to a file (overwrites if it exists).

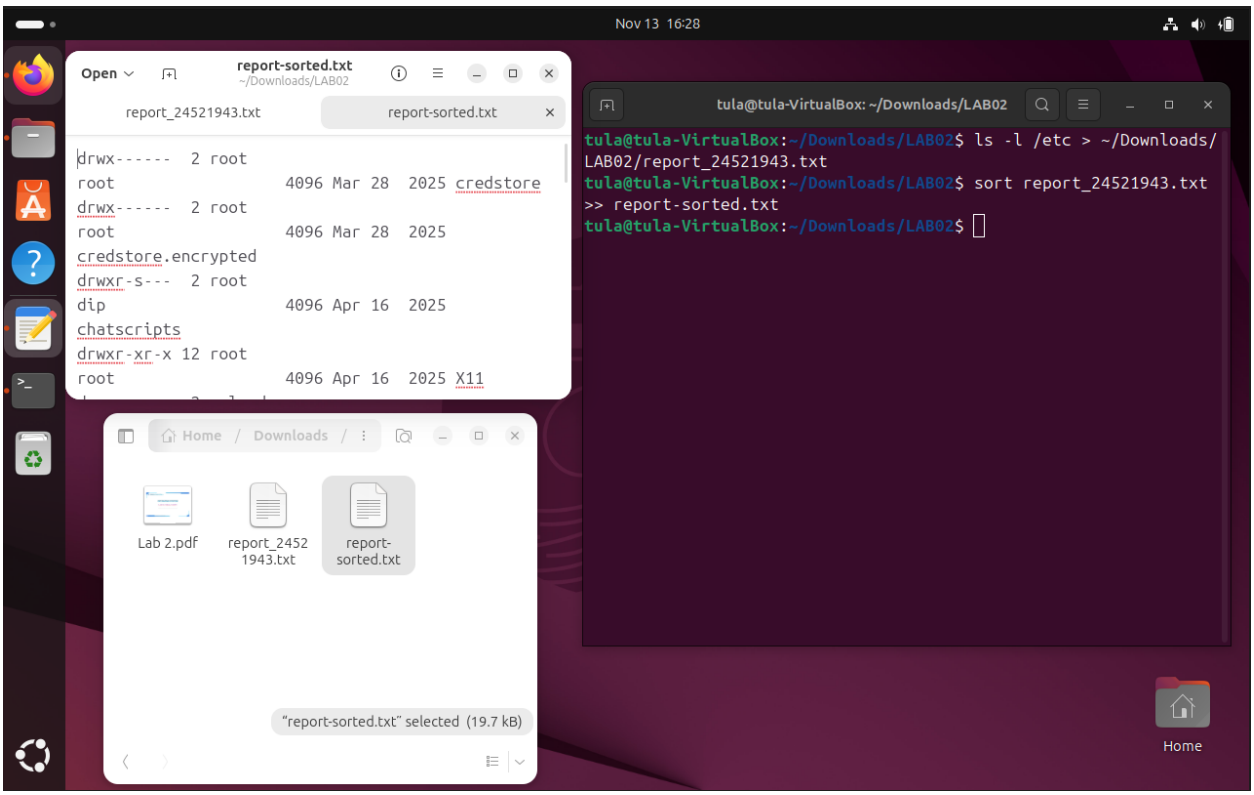
`~/report_23560354.txt`

Stores the report in your **home directory**

2. Sort the contents of the report_<studentID>.txt file from your current directory and append it to the end of a new file named report-sorted.txt.

Answer:

Code: sort report_24521943.txt >> report-sorted.txt



Explanation:

`sort report_23560354.txt`

Sorts the contents of the file alphabetically (line by line).

`>> report-sorted.txt`

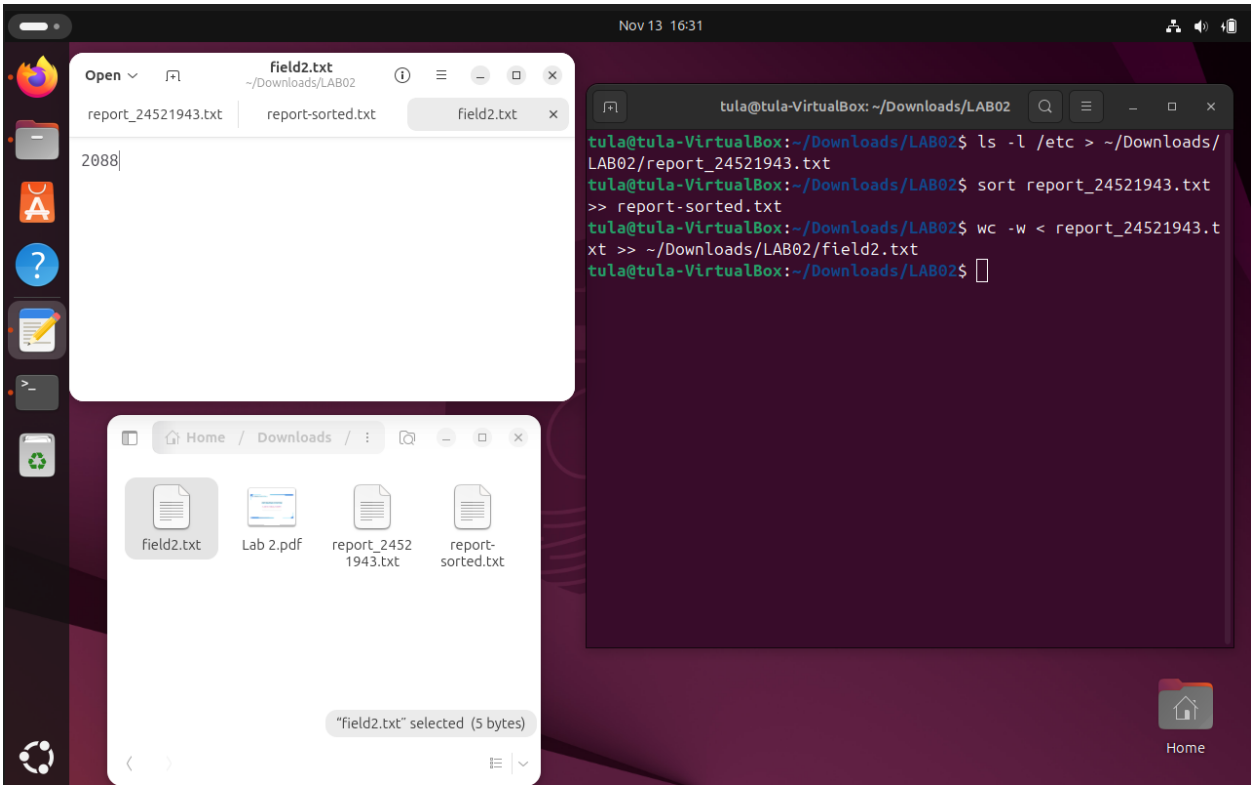
Appends the sorted output to the end of `report-sorted.txt`.

If the file doesn't exist, it will be created.

3. Count the number of words within the report_<studentID>.txt file and *append* the output to the end of a file field2.txt in your home directory. You will need to use both input and output redirection.

Answer:

Code: wc -w < report_24521943.txt >> ~/field2.txt



Explanation:

`wc -w`

Counts the number of **words** in the file.

`< report_23560354.txt`

This is **input redirection** — it sends the file content *into* the `wc` command.

`>> ~/field2.txt`

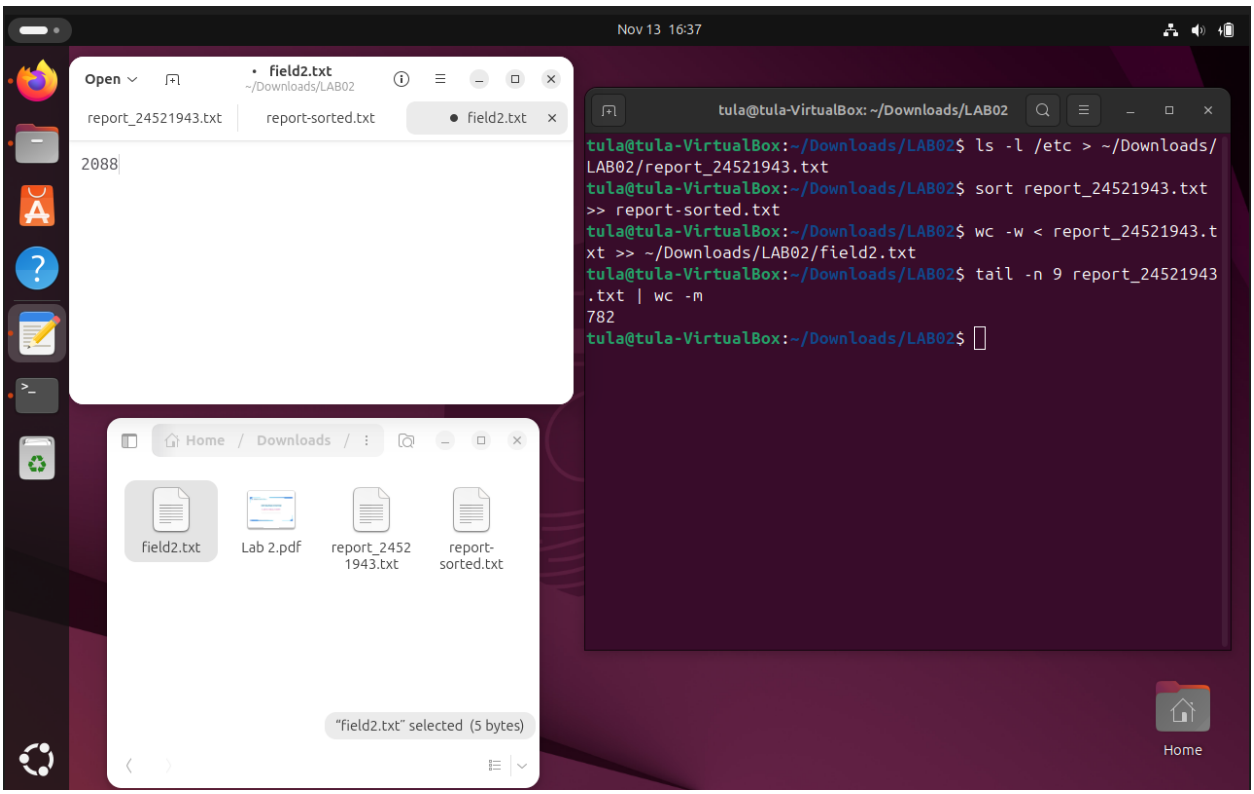
This is **output redirection** — it **appends** the result to the end of `field2.txt` in your home directory.

If `field2.txt` does not exist, it will be created.

4. Using the previously created report_<studentID>.txt file, count the number of characters of the last 9 lines.

Answer:

Code: tail -n 9 report_24521943.txt | wc -m



Explanation:

`tail -n 9`

Extracts the **last 9 lines** of the file.

`wc -m`

Counts the **number of characters**, including spaces and punctuation.

The output will be a single number — the total character count across those 9 lines.